Response to HHS RFI on AI for Clinical Care from SKAF

https://public-inspection.federalregister.gov/2025-23641.pdf

Response to Request for Information: Accelerating the Adoption and Use of Artificial Intelligence as part of Clinical Care (HHS Health Sector AI RFI)

Regarding feedback on how current HHS **regulations** impact AI adoption and use for clinical care:

There is an FDA guidance document for Clinical Decision Support Software (January 2026, FDA-2017-D-6569) that explains certain CDS software functions are "excluded from the definition of device" and thus excluded from regulation enforcement if they meet all 4 of specific criteria, including criterion "(4) intended for the purpose of enabling such health care professional to independently review the basis for such recommendations that such software presents so that it is not the intent that such health care professional rely primarily on any of such recommendations to make a clinical decision or treatment decision regarding an individual patient"

In simple English, a developer of CDS software is motivated to avoid regulation of their CDS software by providing clear rationale/evidence for the recommendation offered by the CDS such that the health care professional can review the rationale/evidence and independently decide on the recommended action. Regulation of CDS software is thus limited to situations where the CDS software is *replacing* clinical judgment without an opportunity to *apply* clinical judgment.

When the recommendation is determined by traditional human efforts, there are traditional forms for explaining the rationale/evidence behind the recommendation, and these forms ideally represent a balanced view of the benefits and harms (i.e. a systematic review of the evidence) rather than a selective view that only reports the evidence that is most supportive of the recommendation. Such forms often include the certainty of or confidence in the evidence and other judgments related to the balance of benefits and harms. Some representations of these forms have been called evidence-based medicine (EBM), ideally used to represent an 'answer' based on systematic and comprehensive evaluation of the evidence rather than selective reporting of 'evidence' to justify a desired 'answer'.

When the recommendation is generated by AI, there is a need to represent the actual rationale/evidence used by the AI and full consideration of the relevant data inputs RATHER THAN selective reporting to justify the recommendation by the AI. There is a high risk of bias if the rationale/evidence reported for 'independent review' by the health care professional is selected AFTER the answer is determined by the AI rather than through a process of recognizing the data that contributes directly to the AI's decision making. In other words, generative AI can generate rationales to support its “answer” that are contextually incorrect. By providing selective evidence for the healthcare professional to review, the review is no longer independent and the AI can bias clinical care.

A method of reporting the evidence supporting an AI-generated recommendation needs to be human-interpretable (to meet the FDA guidance) AND needs to be machine-interpretable (to enable accurate generation and representation by AI). A method that is interoperable and re-usable would go a long way to help everyone across the ecosystem achieve this optimal state of communication.

We have developed an interoperable and re-usable method. This method builds on the technical standard for data exchange for health data, known as the HL7 FHIR standard, or Fast Healthcare Interoperability Resources (FHIR). We have expanded FHIR to support data exchange for science, and this is described in an Evidence Based Medicine on FHIR (EBMonFHIR) Implementation Guide at https://build.fhir.org/ig/HL7/ebm.

The FHIR Resources for Evidence Based Medicine Knowledge Assets ('EBMonFHIR') Implementation Guide defines profiles and value sets for the representation of scientific knowledge. This implementation guide is intended for developers of systems using FHIR for data exchange of scientific knowledge and for authors of more specialized implementation guides in this domain.

This implementation guide covers the broad scope of representation of scientific knowledge, including (1) citations to represent identification, location, classification, and attribution for knowledge artifacts; (2) components of research study design including study eligibility criteria (cohort definitions) and endpoint analysis plans; (3) research results including the statistic findings, definition of variables for which those findings apply, and the certainty of these findings; (4) assessments of research results; (5) aggregation and synthesis of research results; (6) judgments regarding evidence syntheses and contextual factors related to recommendations; (7) recommendations; and (8) compositions of combinations of these types of knowledge. The types of interoperability covered include syntactic (Resource StructureDefinitions) and semantic (value sets).

The EBMonFHIR Implementation Guide includes about 100 Profiles, Extensions, and ValueSets to guide many aspects of data exchange for scientific knowledge.

Regarding feedback on ways in which HHS may **invest in research & development** to integrate AI in care delivery and create new, long-term market opportunities that improve the health and wellbeing of all Americans:

There are many research & development opportunities that can be considered across the entire scope of the EBMonFHIR Implementation Guide.

One specific opportunity to highlight here (for substantial opportunity to improve CDS adoption for specific clinical decisions to improve health and wellbeing) is the SummaryOfNetEffect Profile. The SummaryOfNetEffect Profile is a structure definition for combining all of the following data elements related to a decision:

1. Specification of the Population represented for that decision, e.g. people who have chronic kidney disease, have anemia (low red blood cell count), and are being treated with dialysis
2. Specification of the Intervention represented for that decision, e.g. erythropoiesis-stimulating agents (ESA), which are medicines to stimulate the body to produce more red blood cells
3. Specification of the Comparator represented for that decision, e.g. no ESAs
4. Specification of the outcomes to be considered for that decision, e.g. Needing a blood transfusion, Mortality, and Vascular access thrombosis.
5. For each outcome, specifying:
    a. The risk for the outcome without treatment (or with the Comparator), which could be derived from evidence from clinical research or risk prediction defined for an individual patient
    b. The estimated change in the outcome resulting from treatment (use of the Intervention), which should be derived from evidence from clinical research
    c. The desirability of the outcome (desirable or undesirable), which determines whether the estimated change in the outcome resulting from treatment is a benefit or harm
    d. The relative importance of the outcome, which could be derived from evidence from population surveys or individualized for personal decision making
    e. A net effect contribution (calculated by multiplying the effect estimate and the relative importance)
6. Specification of a net effect estimate (statistically combining the net effect contributions)

An example for a SummaryOfNetEffect for ESAs for CKD patients with anemia receiving dialysis can be seen at https://fevir.net/resources/Composition/422055 -- this is evidence that can easily be shown in human-interpretable and machine-interpretable forms.

A further representation of this evidence, designed to support individualized shared decision making, can be seen at https://netbenefitcalculator.com/422055

There are nearly infinite combinations of how this evidence could be shared to support CDS. But if HHS could invest in research & development to establish and support a common basis for sharing this data, that would create new, long-term market opportunities.

Navigation software (MapQuest, Google Maps) is far advanced today and used in nearly infinite ways because there are common expectations in data sharing with both human-interpretable and machine-interpretable interfaces. But there is currently no standardized navigation system ("GPS") for healthcare.